

3(a)(i) In Fig. 3.1, the electron moves in a projectile motion and trace a parabolic path towards the higher potential plate. B1

There is a resultant force in the vertical direction. There is an acceleration in the vertical direction and a constant velocity in the horizontal direction. B1

Comments

Some students stated that the electron curves or is deflected. This answer is not sufficient. Students need to state parabolic path upwards due to constant acceleration in vertical direction and constant velocity in horizontal direction. The criteria for projectile motion was missing in most scripts.

(a)(ii)

$$\begin{aligned}
 a &= \frac{F}{m} = \frac{qE}{m} = \frac{q\left(\frac{V}{d}\right)}{m} \\
 &= \frac{(1.6 \times 10^{-19})\left(\frac{50}{5 \times 10^{-3}}\right)}{9.11 \times 10^{-31}} \\
 &= 1.8 \times 10^{15} \text{ ms}^{-1}
 \end{aligned}$$

M1

A1

(a)(iii) Time taken for electron to move between the plates =

$$\frac{s_x}{u_x} = \frac{9 \times 10^{-3}}{1.5 \times 10^7} = 6.0 \times 10^{-10} \text{ s}$$

M1

Vertical speed of electron when it emerges from plates,

$$v_y = u_y + at$$

$$v_y = 0 + (1.8 \times 10^{15})(6 \times 10^{-10}) = 1.08 \times 10^6 \text{ ms}^{-1}$$

M1

$$\tan \theta = \frac{v_y}{v_x} = \frac{1.08 \times 10^6}{1.5 \times 10^7}$$

$$\theta = 4.1^\circ$$

Comments

(a)(ii) was well-attempted. In a(iii), some students erroneously used the ratio of the displacements to obtain the angle of deflection, which is incorrect. Students should use the ratio of velocities.

A1

(b)(i) In Fig. 3.2, the electron moves in an oscillatory motion about the line that joins the centre of masses of the positive charges. It moves to and fro along the horizontal line and is always equidistant between the two positive charges.

The resultant force is horizontal and always points towards the centre of masses of the two positive charges. Hence, the acceleration is in the horizontal direction and points towards the centre of masses of the two positive charges.

Comments

This part was poorly attempted. Students were not able to identify an oscillatory motion.

(b)(ii)

$$F = \frac{Qq}{4\pi\epsilon_0 r^2} = \frac{(1.6 \times 10^{-19})^2}{4\pi\epsilon_0 \left(\sqrt{(5 \times 10^{-3})^2 + (2 \times 10^{-3})^2} \right)^2} = 7.9376 \times 10^{-24} \text{ N}$$

M1

Resultant force = $2F \cos \square$

$$\begin{aligned} &= 2(7.9376 \times 10^{-24}) \frac{5 \times 10^{-3}}{\sqrt{(5 \times 10^{-3})^2 + (2 \times 10^{-3})^2}} \\ &= 1.5 \times 10^{-23} \text{ N} \end{aligned}$$

M1

A1

Comments

Most students showed understanding that the resultant force is in the horizontal direction.